

Conceptual Questions: Order Flow Imbalance

1. What’s the motivation behind measuring OFI at multiple depth levels of the order book?

The motivation for measuring OFI (Order Flow Imbalance) at multiple depth levels stems from the fact that order book dynamics are not limited to just the best bid and ask prices (Level 0). While the best level reflects immediate trading interest, deeper levels (e.g., Levels 1–10) capture latent liquidity and strategic behavior of market participants.

The research paper *Cross-Impact of Order Flow Imbalance in Equity Markets* shows that multi-level OFI vectors provide significantly greater explanatory power for return dynamics compared to best-level OFI alone. Deeper-level orders can affect price formation, especially when market conditions change or large orders interact with the book.

In implementation of the task, I computed raw OFIs from Levels 1 to 10, capturing a more complete view of the book. When I applied PCA to compress these into an integrated OFI, we found that deeper levels contributed meaningful weights, validating their informational value.

2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors opt for Lasso regression instead of Ordinary Least Squares (OLS) when estimating cross-asset OFI impact due to the high dimensionality of the problem. With hundreds of assets, a cross-impact model includes a large number of regressors—one for each asset’s OFI.

OLS is prone to overfitting, multicollinearity, and instability in such settings. Lasso regression, which adds an ℓ_1 regularization term, promotes sparsity in the coefficient vector. This means it selects only the most relevant cross-asset OFI predictors, effectively shrinking the rest to zero. Additionally, Lasso improves model interpretability by selecting only the most impactful predictors, which is especially important in financial applications where understanding the role of each input is critical for transparency and risk management.

3. Why is OFI considered a better predictor of short-term returns than trade volume?

Trade volume alone does not contain directional information—it simply captures the total size of trades. In contrast, OFI captures the imbalance between buy and sell interest by accounting for how bid and ask sizes evolve relative to price changes.

The paper highlights that returns respond more to shifts in order book imbalance than to raw volume. A positive OFI suggests buy-side dominance, which often leads to upward price movement, while a negative OFI signals sell-side pressure.

In my analysis while performing the task, I observed that spikes in best-level and integrated OFI aligned with subsequent returns. This directional insight, absent in volume data, makes OFI a superior predictor of short-term price changes.